



ECA Project 1

Group 8
Mansi Bhardwaj, Thomas Kas, Jeffrey van Velzen



Hardware Descriptions

- Base: ISA: I X
- Medium: ISA: C I M X
- Performance: ISA: C I M X, FAST_MUL

I: Integer base ISA (always enabled)

X: NEORV32-specific extension (always enabled)

C: Compressed instructions

M: Integer multiplication/division instructions

All at 100MHz

Bottlenecks: load/store wait cycles (cache-misses, access memory more linearly), load instructions, instruction dispatch wait cycles/branch instructions (compiler flags), ALU wait cycles (data dependencies, compiler flags/order)



Optimizations

- Code optimizations:
 - Pointers used for each row
 - Constant substitution for kernel values
 - Temporary sum variable with manual kernel unrolling
- Compiler optimizations
 - Optimization levels: -O0, -O1
 - Compiler flags:
 - -funroll-loops
 - -fomit-frame-pointer
 - -fno-tree-vectorize
 - -fcse-follow-jumps
- Best combination: -O1 and all compiler flags



Performance Results

Configuration	Initial state	Code optimization	Code + compiler optimization
Base	571,898 cycles	38,086 cycles	7,447 cycles
Medium	571,892 cycles	38,494 cycles	7,447 cycles
Performance	571,900 cycles	37,750 cycles	7,447 cycles



Energy-Delay Product Results

$$\text{EDP} \approx |\text{LUTs}| \times (|\text{cycles}|/\text{frequency})^2$$

- $\text{EDP}_{\text{base}} \approx 2308 \times (7447/1000000000)^2 \approx 1.28 \times 10^{-5}$
- $\text{EDP}_{\text{medium}} \approx 4868 \times (7447/1000000000)^2 \approx 2.70 \times 10^{-5}$
- $\text{EDP}_{\text{performance}} \approx 6958 \times (7447/1000000000)^2 \approx 3.86 \times 10^{-5}$

Our winner: base.

GPU Acceleration of 3×3 Convolution

ECA PROJECT 2

GROUP 8

Mansi Bhardwaj, Jeffrey van Velzen

Platform Choice & Specifications



**Chosen Platform: NVIDIA GPU
with CUDA**



Key Specifications:

Parallel processing architecture
Thousands of concurrent threads
High memory bandwidth
Shared memory hierarchy

Problem Overview

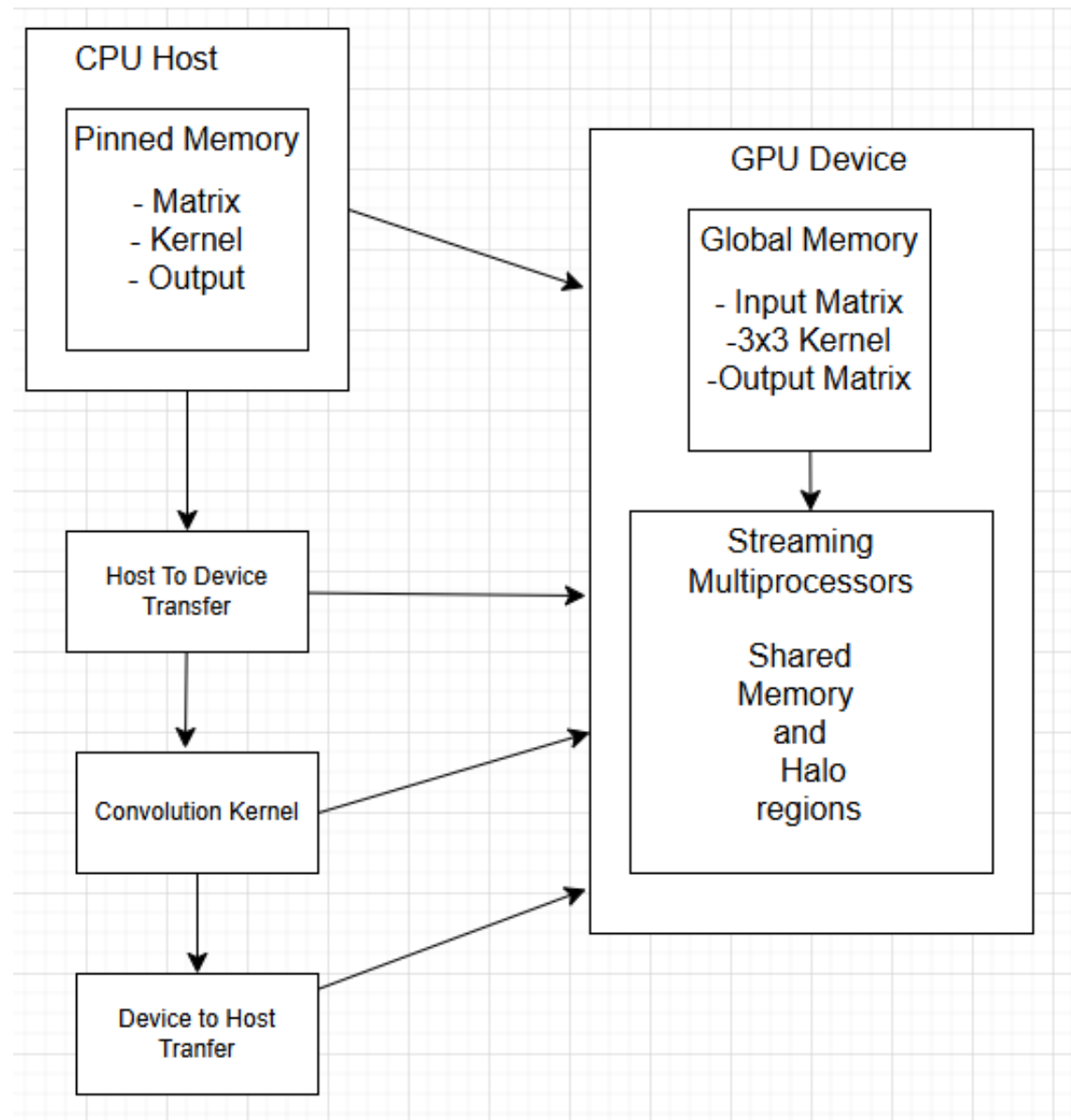
Operation: 3×3 Convolution on 1300×1300 matrix

Data Type: 32-bit signed integers

Challenge: Process 1.69 million output elements

Visual: Show convolution operation with 3×3 kernel sliding over large matrix

GPU Execution Block Diagram

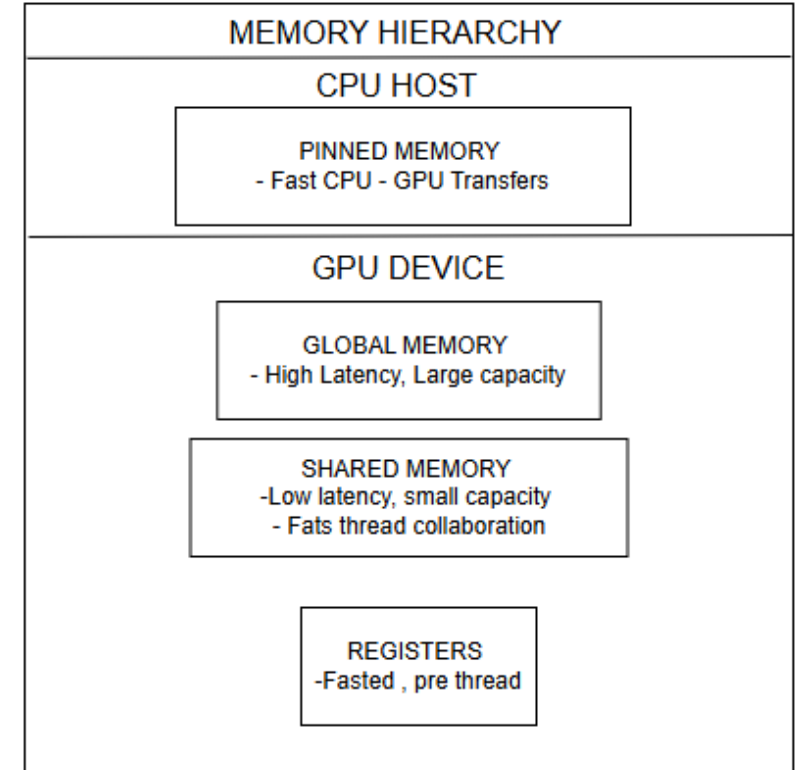


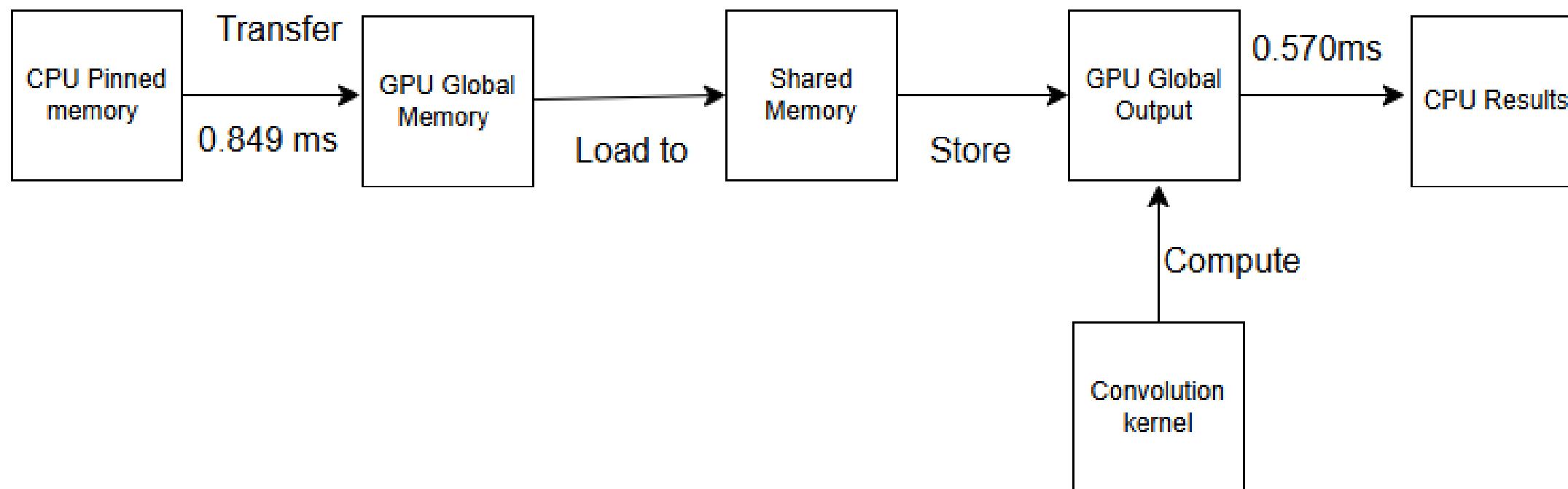
Memory Optimization

- **Pinned Memory:** Faster CPU-GPU transfers
- **Shared Memory:** Reduces global memory accesses
- **Halo Regions:** 16×16 tile + 1-pixel border = 18×18 shared memory

Optimization Benefits

- **Shared Memory vs Global Memory:**
 - Shared: 1.050 ms (coalesced access pattern)
 - Global: 0.106 ms (but more memory bottlenecks)
- **Key Insight:** Memory transfers dominate execution time
- **Speedup:** $\sim 11 \times$ faster than CPU implementation





Data Flow Diagram

Performance Results

```
Matrix size: 1300x1300, Kernel size: 3x3  
CPU time: 32.083 ms
```

```
--- GPU Timing Breakdown (Optimized) ---  
Host → Device transfer: 0.849 ms  
Kernel execution:      1.050 ms  
Device → Host transfer: 0.570 ms  
Total GPU time:        2.837 ms  
Global memory kernel:  0.106 ms  
GPU matches CPU results!
```